



Inter-Satellite Link Prediction with Supervised Learning Based on Kepler and SGP4 Orbits

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Received: 14 November 2023 / Accepted: 18 July 2024
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Abstract

Distributed Space Systems (DSS) are gaining prominence in the space industry due to their ability to increase mission performance by allowing cooperation and resource sharing between multiple satellites. In DSS where communication between heterogeneous satellites is necessary, achieving autonomous cooperation while minimizing energy consumption is a critical requirement, particularly in sparse constellations with nano-satellites. In order to minimize the functioning time and energy consumed by the Inter-Satellite Links established for satellite-to-satellite communication, their temporal encounters must be anticipated. This work proposes an autonomous solution based on Supervised Learning that allows heterogeneous satellites in circular polar Low-Earth Orbits to predict their close-approach encounters given the Orbital Elements. The model performance is evaluated and compared in two different scenarios: (1) a simplified scenario assuming Kepler orbits and (2) a realistic scenario assuming Simplified General Perturbations 4 orbital model. The obtained results demonstrate a Balanced Accuracy exceeding 95% when compared to realistic data from an available database. This work represents a promising initial stage in developing an alternative approach within the field of DSS.

Keywords Distributed space systems · Non-terrestrial networks · Sixth-generation (6G) wireless networks · Inter-satellite links · Low-earth orbits · Kepler · Simplified general perturbations · Simplified perturbations models · Supervised learning

1 Introduction

In recent years, the space industry has undergone a significant shift toward the implementation of Distributed Space Systems (DSS) [1]. As opposed to traditional monolithic systems, DSS consists of multiple satellites that communicate and coordinate their activities to increase the overall mission

performance and provide global and continuous coverage for applications such as telecommunications, navigation, and remote sensing.

The growing interest in Non-Terrestrial Networks (NTNs), a fundamental concept in sixth-generation (6G) wireless networks, and the increasing number of launched satellites and constellations emphasize the importance of real-time satellite-to-satellite communication through Inter-Satellite Links (ISLs). ISLs in DSS can establish an optimal cooperation plan to further increase the mission performance and meet strict requirements, particularly crucial in scenarios with limited resources such as nano-satellite constellations.

Due to the dynamic nature of satellites operating in heterogeneous environments with varying altitudes and inclinations, ISLs experience intermittent connectivity and have limited lifetimes. Anticipating the time intervals during which two satellites are in close proximity, known as encounters, is essential. This anticipation not only conserves energy by preventing failed communication attempts but also enables the creation of an optimal satellite communication plan.

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The traditional solution for satellite encounter prediction is based on deterministic and centralized on-ground orbit propagation [2, 3]. This approach utilizes an orbital model to propagate the satellite's initial state, which is encoded in the Two-line Elements (TLE). Given the satellite trajectories, close approach encounters can be defined from a maximum satellite-to-satellite distance threshold for communication. While on-ground propagation solutions can generate precise contact plans, their centralized nature makes them unsuitable for DSS with heterogeneous satellites, as substantial coordination efforts between different stakeholders would be required. Additionally, the scalability of centralized computations is limited due to the extensive communication via ground links and the significant centralized processing required to manage data for mega-constellations.

To overcome these limitations, new investigations are moving towards deterministic and decentralized solutions, providing autonomy to the satellites [4]. However, these techniques are still based on orbital propagation, which results in precise solutions but can result in a high processing cost for satellites with limited resources, such as nano-satellites. To avoid orbital propagation while keeping a decentralized and autonomous solution, authors in [5] have designed predictive algorithms to self-learn and self-construct the contact plan. However, in order to mathematically formulate the solution, they apply some linearizations only valid for circular and same-high satellites.

Several studies have investigated the use of Graph Neural Networks (GNNs) together with Recurrent Neural Networks (RNNs) to model the temporal evolution of dynamic systems and predict future behavior based on past data [6–12]. The feasibility of treating a constellation as a time-evolving graph has been studied in [13] but discarded due to its complexity, poor performance, and strong dependency on past ground-truth data. However, new studies [14] are further investigating in this promising direction.

Additionally, recent studies are exploring satellite collaboration techniques using decentralized reinforcement learning to find optimal routing paths between satellites [15–19]. Typically, the goal is to minimize end-to-end delay while avoiding congestion. However, these studies assume homogeneous environments where all satellites share the same altitude and inclination, forming a mesh-like topology. In this topology, each node has four direct neighbors connected through four ISL antennas. While this setup benefits from static ISLs that eliminate the need for encounter prediction, it fails to fully exploit the potential communication resources available in more dynamic and heterogeneous environments.

In summary, current solutions for satellite cooperation present notable strengths in providing precise contact plans through centralized orbit propagation, offering autonomy with decentralized solutions, and exploring innovative predictive algorithms based on Artificial Intelligence. However,

these approaches face challenges such as high processing costs and scalability, limitations due to linearization assumptions in orbital dynamics, and simplifications with homogeneous satellite environments. Addressing these challenges will be crucial for enhancing the scalability, adaptability, and effectiveness of future distributed space systems and NTN, paving the way for more robust and efficient satellite operations in diverse and dynamic orbital environments.

Building on the previous publication [20], the current work presents some advances in the proof of concept of using supervised learning to predict the encounters between heterogeneous satellites. Concretely, this work presents and compares the results of predicting the close approach encounters between polar Low Earth Orbit (LEO) satellites in two different scenarios: (1) ideal scenario assuming that satellites follow Kepler orbits, and (2) realistic scenario assuming that satellites follow Simplified General Perturbations 4 (SGP4) orbital model. While Kepler orbits follow a simplified model considering only the gravity attraction between two bodies, Simplified SGP4 is a well-known and realistic orbital model specifically used for LEO objects as it accounts for the two most significant perturbations in near-Earth space: atmospheric drag and Earth's non-sphericity [21]. In both scenarios, the same model architecture and input data are used.

Even though the computation cost of the approach is a crucial point, this work focuses on obtaining good prediction performances. The analysis of the inference time and resources is left for future work.

The remainder of the article is structured as follows. Section 2 defines the problem statement, introducing the datasets used for training, validation, and testing. Section 3 details the SL model architecture used to fit the input and output data. Section 4 presents the model performance in the validation and test set and compares the results obtained with the two different orbital models. Finally, Sect. 5 concludes the paper with a summary of keyfindings and suggestions for future research in this area.

2 Problem Statement

The encounter prediction problem entails forecasting communication opportunities between satellites based on their proximity. Encounters occur when the satellite-to-satellite distance falls below a specified threshold (T).

In a satellite constellation with shared altitude and inclination, minimal relative movement is observed. In such scenarios, encounters are either consistently available or unavailable, requiring no anticipation. However, in heterogeneous environments, ISLs exhibit intermittent connectivity and encounter anticipation is required for optimal cooperation. As an illustrative example, Fig. 1a shows the encounters

Table 1 OE ranges for circular, polar, and low orbits

OE	e [–]	M [°]	i [°]	ω [°]	Ω [°]	a [km]
From	0	0	90	0	0	6800
To	0	360	100	0	360	7300

e eccentricity, M mean anomaly, i inclination, ω argument of periapsis, Ω right ascending node, a semi-major axis

for a given pair of satellites with relative movement between them.

The solution proposed in this work involves training a SL model with a large set of input-output pairs. Figure 1b illustrates a simplified diagram of the three main steps that define the solution: data generation, model training, and model evaluation.

The role of the data generation part is to create the input and output data used for training, validation, and testing. On the one hand, the input data contains the Orbital Elements (OE) for different satellite pairs, a set of six parameters that locate a satellite by uniquely identifying its orbit shape, size, and orientation. Both training and validation input data are synthetically generated from 1000 different OE random values within the range for circular, polar, and low orbits defined in Table 1. Notice that, since all orbits are circular, both the eccentricity (e) and the argument of periapsis (ω) are always zero and can be excluded from the input set. As an example, Table 2 presents the first six rows of the input training data, featuring the satellite pairs 0–1, 0–2, 1–0, 1–2, 2–0, and 2–1. Notice that, even though pairs $i-j$ and $j-i$ generate exactly the same output, the input vectors are different, and therefore, both must be included in the training set.

On the other hand, the output ground-truth data is the binary time sequence that determines when the satellite-to-satellite distance is below a given distance threshold. A total of 576 binary values are used to characterize each output, corresponding to 48 h of encounter discretized into intervals of 5 min.

Note that these encounters are directly influenced by the input data, the orbital model used to compute the satellite trajectories, and the distance threshold. The last, has a huge impact on the number and the duration of communication opportunities: higher thresholds result in longer encounter duration, while lower ones lead to more sporadic encounters. The specific antenna characteristics of each satellite, including factors such as power, antenna gain, radiation pattern, and pointing direction, directly impact its maximum coverage distance. In this work, it is assumed that the antenna pointing process is done automatically and that the other characteristics allow for a maximum distance cover of 2000 km. However, it is important to clarify that our proposed solution can be adapted to any specific antenna characteristics simply by creating the corresponding training output dataset.

Regarding the orbital model, both Kepler and SGP4 orbit propagators are used and compared. Kepler orbit determination relies solely on the six OE mentioned previously. In contrast, SGP4 requires an additional drag term to account for the atmospheric drag perturbation. A range of common drag values is obtained by analyzing the Two-Line Element (TLE) set of various real polar LEO satellites. Prior to the SGP4 propagation, each synthetic satellite is assigned a distinct drag term. Consequently, slightly different ground truth encounters are obtained with Kepler and SGP4 orbital models.

In the second part of the diagram in Fig. 1b, the SL model is fitted with the training input-output sequences: 95% of synthetic samples are used for training, while the remaining part is used for validation to avoid overfitting. Details about the model architecture are presented in Sect. 3.

Finally, the model parameters learned during training are saved and used to predict both validation and test sets. The test set is a collection of 48 realistic polar LEO satellites obtained from Celestrak [22]. The model performance, presented in Sect. 4, is obtained by comparing the predicted encounters with the ground truth.

Note that, to anticipate the encounter, this approach assumes the satellite requires not only the trained model and its own OE values but also the OE values of the other satellite. Additionally, both sets of OEs must be defined at the exact same time, coinciding with the start of the encounter signal. This information is assumed to be known with enough anticipation.

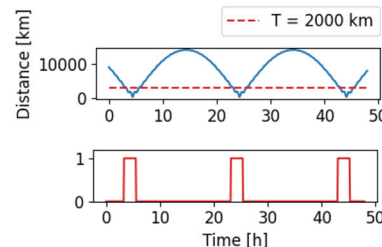
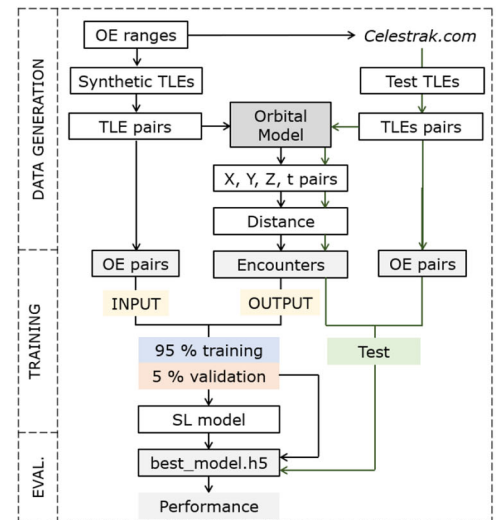
It is important to note that our approach remains effective even when satellite pairs have varying antenna capabilities. Assume each satellite is equipped with its own encounter predictive SL model tailored to its specific antenna range. If a transmitting satellite detects an encounter and initiates communication, the receiving satellite may not respond immediately (excluding propagation delay) due to its antenna limitations. However, the acknowledgment message will be sent after a few minutes once the separation distance is sufficiently small for the second satellite to transmit.

3 Supervised Learning Model

Supervised Learning (SL) is a well-known machine-learning method used to solve problems where a large set of input-output pairs can be obtained yet the mathematical relationship between them is highly complex, costly, or unknown. While the training stage is usually costly, the simplicity of the operations in the inference part renders it a cost-effective approach, which could be suitable for resource-constrained systems like nano-satellites. However, it is important to clarify that this work focuses on finding the SL model architecture that best fits the input data. Future research will

Table 2 First six samples of the input training set, containing the orbital necessary information for satellites 0, 1 and 2

Pair ID	$M1$ [°]	$i1$ [°]	$\Omega1$ [°]	$a1$ [km]	$M2$ [°]	$i2$ [°]	$\Omega2$ [°]	$a2$ [km]
0–1	252.1	95.2	346.6	7112	174.2	97.7	165.0	6906
0–2	252.1	95.2	346.6	7112	178.2	96.0	257.8	6746
1–0	174.2	97.7	165.0	6906	252.1	95.2	346.6	7112
1–2	174.2	97.7	165.0	6906	178.2	96.0	257.8	6746
2–0	178.2	96.0	257.8	6746	252.1	95.2	346.6	7112
2–1	178.2	96.0	257.8	6746	174.2	97.7	165.0	6906

Fig. 1 Block diagram and encounter definition(a) Encounter defined as the time steps at which the satellite-to-satellite distance is below a given distance threshold (T).

(b) Block diagram showing the most important blocks used for data generation, model training and model evaluation

address tasks concerning resource optimization and trade-off analysis between performance and computational costs.

Figure 2 illustrates the Fully-Connected Neural Network (FCNN) architecture proposed to map the input OE pairs to the output sequence of encounters.

FCNN stems from its capability to learn complex nonlinear relationships between inputs and outputs, which is crucial in tasks requiring high-dimensional mappings like ours.

As stated in Sect. 2, the input set consists of 8 values, while the output describes the next 48 hours of encounters with 576 binary features. Since the number of input features is much lower than the number of output features, increasing the number of neurons while progressing deeper into the model is a good approach.

Additionally, the number of neurons per layer is often chosen to be a power of two, mainly for reasons of hardware efficiency, memory management, and computational performance.

Although the optimal number of layers and neurons per layer depends on the complexity of the prediction tasks and on the number of input and output features, there is no direct formula to determine the optimal model architecture. Instead,

it involves a trial-and-error process, starting with simple architectures (with few trainable parameters) and progressively increasing complexity while avoiding overfitting.

Based on this reasoning, multiple architectures with varying numbers of hidden layers and neurons were evaluated. Among all the evaluated models, the one depicted in Fig. 2, with 6 hidden layers and 270,048 trainable parameters, shows one of the best training curves with high generalization. It has been observed that the strong imbalance between zeros and ones present in the output training data causes less complex SL architectures to struggle in predicting positive encounters.

Additionally, other important considerations have been taken into account to mitigate potential training issues arising from the imbalanced training set. The Rectified Linear Unit (ReLU) is used as the activation function for the first seven layers (excluding the output layer). ReLU's ability to maintain stronger gradients during backpropagation helps improve learning efficiency and model performance. Similarly, the Sigmoid activation function has been applied at the output layer to further simplify the training process by ensuring each predicted output value falls within the range

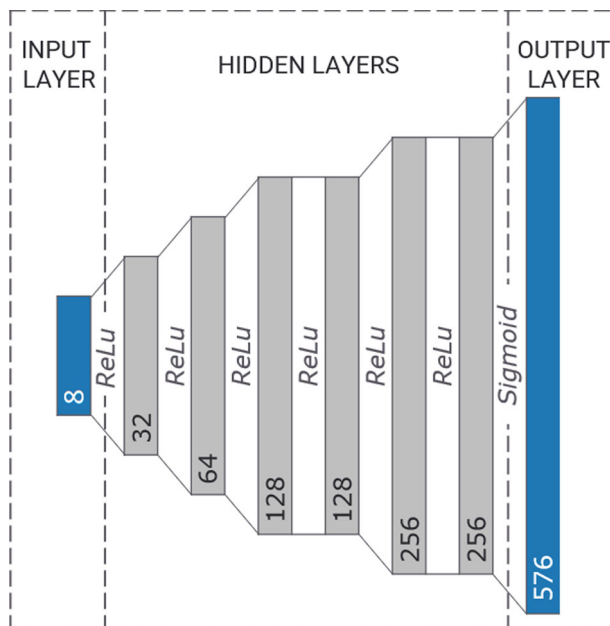


Fig. 2 SL model architecture consisting of 8 layers. The values within each layer represent the number of neurons while the word in-between each layer is the applied activation function

from 0 to 1. In that sense, a decision threshold should be used to post-process the model output and decide if the predicted values should be considered as no encounter (values close to 0) or encounter (values close to 1).

Additionally, the binary cross-entropy loss function is chosen for its effectiveness in binary classification tasks, ensuring the model correctly distinguishes between the two classes [23]. The Adam optimizer is selected for its adaptive learning rate and efficient handling of sparse gradients, leading to faster convergence [24]. Early stopping with a patience of 200 epochs prevents overfitting by halting training when performance ceases to improve, ensuring the model generalizes well to unseen data [25].

Other important training settings include 500 epochs of training with a batch size of 3000 samples.

4 Results

This section presents the performance of the SL model introduced in Sect. 3. Two scenarios are evaluated and compared: the ideal scenario and the realistic scenario. In the ideal scenario, the output sequence of encounters used for training, validation, and testing is computed using Keplerian orbits, an ideal orbital model that neglects external perturbations. In contrast, the realistic scenario assumes SGP4 model, which incorporates Earth's shape and atmospheric drag perturbations. Consequently, two distinct SL models are trained with the same input data but different output data. Poorer results

are expected from the SGP4-based model due to the absence of output-dependent information concerning the initial drag coefficient.

Figure 3 presents a comparison between the encounter ground truth (in blue) and the model prediction (in orange) for three different test satellite pairs during a specific time range of interest and for both Kepler and SGP4 trained models, respectively. At first sight, the results show that both models can correctly predict the three encounters coming from unseen and realistic input data, even in complex scenarios where contacts are short in time. No major difference can be observed between Kepler and SGP4 results.

It should be considered that encounters are usually rare, especially in sparse heterogeneous constellations. In particular, for the set of 1000 synthetic polar LEO satellites, the average time a pair of satellites is within 2000 km of proximity is only 2%. This implies that the output ground truth data is expected to have around 98% zeros. In such highly unbalanced scenarios, using accuracy to evaluate the results is misleading, as an untrained model predicting all zeros would show a high performance of around 98%.

Instead, a common metric used in unbalanced binary classification tasks is the F1 score ($F1$). Equation (1) defines $F1$ as the harmonic mean of Precision (P) and Recall (R). Precision measures the error from ones predicted as zero (missed encounters) while Recall measures the errors from zeros predicted as ones (false encounters). Therefore, in our scenario, P can be seen as the probability of estimating a contact inside the real timeslot (reflecting the wasted energy in predictions), while R assesses the proportion of successful encounters relative to the total available communication time (indicating missed communication opportunities).

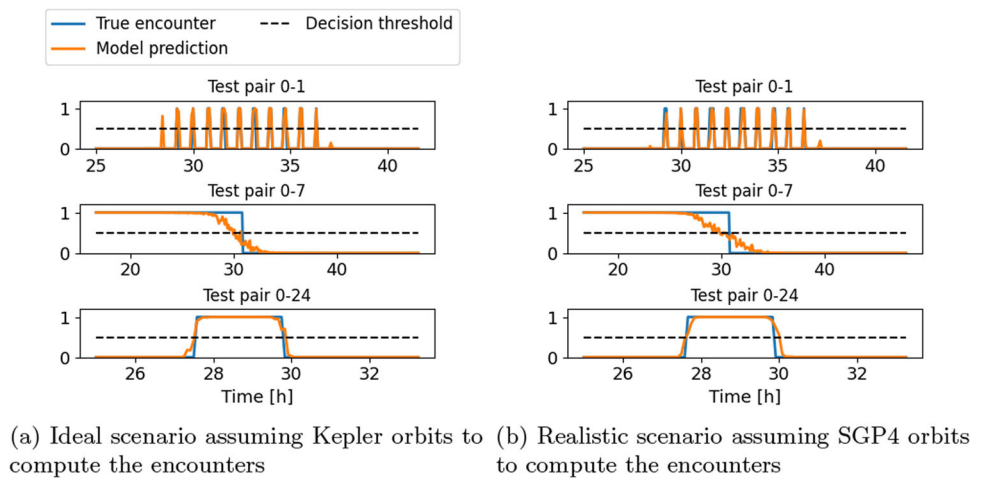
Therefore, the $F1$ score is not only useful for evaluating the unbalanced dataset but also provides two tangible and easily understood metrics that can serve as a performance lower bound depending on mission requirements.

$$F1 = \frac{2 \times P \times R}{P + R} = 2 \times \frac{\frac{TP}{TP+FP} \times \frac{TP}{TP+FN}}{\frac{TP}{TP+FP} + \frac{TP}{TP+FN}} = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (1)$$

where TP, FP, and FN stand for True Positive, False Positive, and False Negative, respectively.

Additionally, the Balanced Accuracy (BA) is specifically designed for unbalanced datasets, as it computes the mean accuracy considering each class. For a binary dataset, BA can be calculated as the mean of Recall and Specificity (see Eq. 2). While Recall has been defined previously, Specificity, also called True Positive Rate (TPR), indicates the model's effectiveness in correctly predicting negative instances, providing an alternative measure of the wasted energy in predictions.

Fig. 3 Comparison of the true encounters (in blue) with the model prediction (in orange)



Note that, BA is especially useful in critical situations where no encounters are predicted, as unlike $F1$, its performance does not drop to zero when $TP = 0$ but accounts for the accuracy of correctly predicting the negative encounters.

$$BA = \frac{R + TNR}{2} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (2)$$

Note that a decision threshold needs to be applied at the output of the model to compute TP, FP, and FN. The decision threshold plays an important role in the performance of the model and should be used depending on the user interests: if the maximum number of encounters want to be detected, the threshold should be reduced; otherwise, if the false encounters want to be minimized, the threshold should be increased. To observe the effect of the decision threshold, Fig. 4 illustrates, for each scenario and dataset, the evolution of the mean performance for all metrics $F1$, P , R , and BA, across all possible decision thresholds from 0 to 1.

As expected, as the decision threshold rises, the count of detected encounters decreases, resulting in a reduction of FPs and an increase in both FNs and TNs. On the one hand, the decrease in FPs enhances P , elevating the likelihood of accurately estimating a contact within the real time slot. However, the increase in FNs negatively affects R , as it diminishes the proportion of successful encounters. A good trade-off between precision and recall is observed to be close to a decision threshold of 0.5, where both curves cross or where the $F1$ -score is maximized. On the other hand, the Balanced Accuracy decreases but not as fast as the recall since it also incorporates the specificity, which increases with the decision threshold. Note that the drastic drop of $F1$ to zero when no True Positives are detected makes BA metric less demanding and consistently higher.

Figure 4 also indicates that Celestrak scenario presents a notably higher area under the curves than the Synthetic

dataset. Concretely, Celestrak shows an area under the curves of 0.06 and 0.03 units higher than the Validation $F1$ and BA curves, respectively. This behavior comes from the distinct characteristics of each dataset. On the one hand, the Synthetic satellites are uniformly distributed, whereas real satellites from Celestrak tend to be densely clustered within specific regions and constellations, resulting in longer satellite encounters and better balance between positive and negative instances. On the other hand, the validation dataset has an inclination range between 90° and 100° , whereas, in a typical satellite constellation, such as the test Celestrak dataset, the polar satellites are confined to a much smaller inclination range between 96° and 100° , corresponding to the Sun-synchronous orbits. Both the higher dataset balance and the lower range of input values facilitate the prediction.

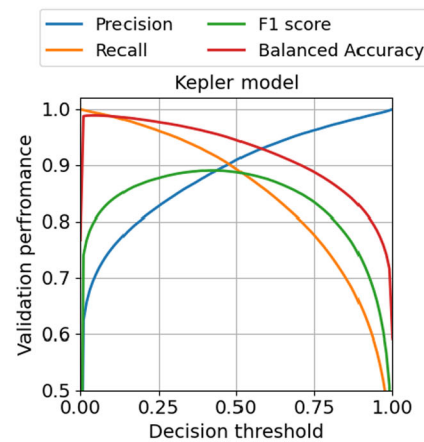
Additionally, the areas under the performance curves also reveal a slightly better performance in the Kepler scenario, with an area 0.02 and 0.01 units higher with $F1$ and BA respectively. To further compare both scenarios and extract conclusions regarding the statistical difference, both $F1$ and BA are computed for each individual sample in both validation and Celestrak sets.

For the following results, a threshold of 0.5 is used, meaning that all model-output values higher or equal to 0.5 are set to one, while values below this threshold are set to zero.

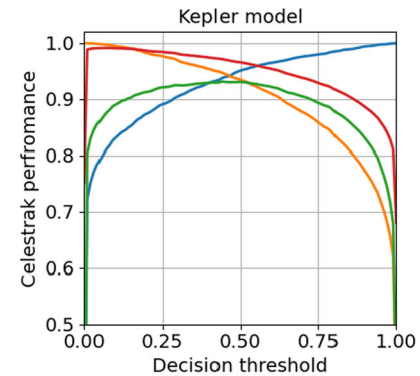
Figure 5 shows the $F1$ score (triangle below the diagonal) and the BA (triangle above the diagonal) for the validation and Celestrak samples in both Kepler and SGP4 scenarios. Only a subset of 1176 validation samples from 49 synthetic satellites has been considered for visualization purposes. The pairs with zero $F1$ have a limited number of encounters, in which having zero TP is almost unavoidable, and a single FP or FN leads to an overall $F1$ of zero. In contrast, BA helps to gain a more comprehensive understanding of the model's performance as it also considers the True Negative Rate.

We conducted a statistical test on the performance of both Kepler and SGP4 models to determine whether a significant

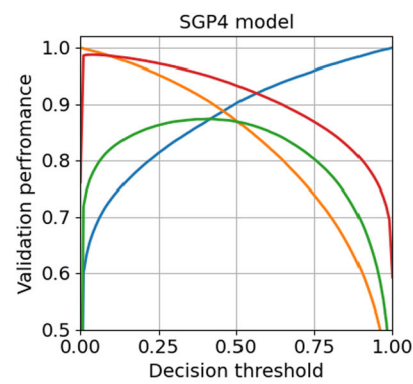
Fig. 4 Decision threshold influence on F1-score, Precision, Recall, and Balanced Accuracy



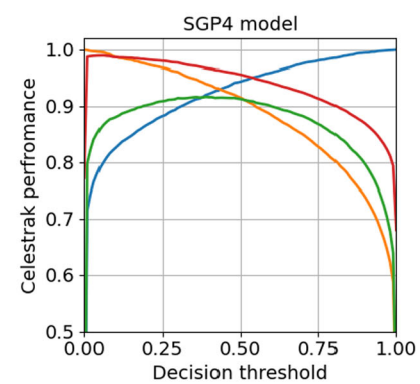
(a) Validation performance of the ideal scenario assuming Kepler orbits.



(b) Celestrak performance of the ideal scenario assuming Kepler orbits.



(c) Validation performance of the realistic scenario assuming SGP4 orbits.



(d) Celestrak performance of the realistic scenario assuming SGP4 orbits.

difference exists between their predictions. As expected, the results confirm a significant difference between the two, with better performance obtained in the ideal scenario.

Table 3 shows the mean BA, as well as the mean $F1$, mean P , and mean R for both scenarios and datasets. Despite output-dependent information missing in the realistic case with SGP4, the mean results for both scenarios are very similar, with BA above 90% and a $F1$ above 85%. This behavior indicates that, despite the stark difference between Keplerian and SGP4 trajectories, the satellite-to-satellite distance, and therefore, the output sequence of encounters are quite similar. Consequently, only knowing the semi-major axis a , the mean anomaly M , the right-ascending node Ω , and the inclination i is enough to anticipate 2 days of encounters with $F1$ and BA of around 95% and 92%, respectively. However, this is only true for small propagation times of less than a week: as time progresses, the influence of missing initial state information related to Earth's shape and drag (epoch, satellite mass, and drag coefficient) gains exponential importance.

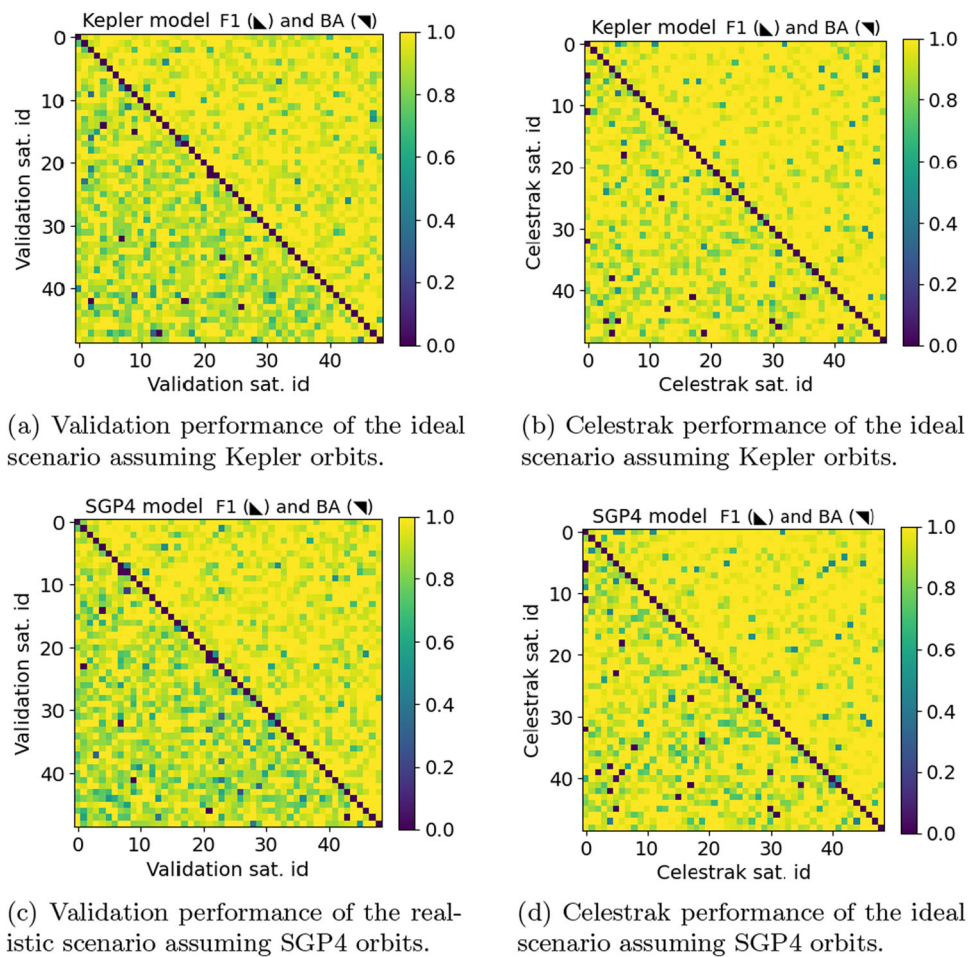
Nevertheless, the statistical test considering the performance of each validation and test samples demonstrates a statistical difference between the two models.

5 Conclusions

This work presents a supervised machine-learning approach for forecasting encounters between satellites in low, circular, and polar orbits in a heterogeneous environment.

The model uses 8-dimensional synthetic input training data obtained by combining various pairs of orbital elements within the region of interest for low, circular, and polar orbits. The output data consists of 48-hour binary time series that describe the encounters of different satellite pairs based on a maximum threshold distance of 2000 km. Two different orbital models are considered to compute the output data used for training, validating, and testing: ideal Kepler model and realistic SGP4, which accounts for the most important perturbations in LEO orbits. A Fully-Connected Neural Network architecture is used for training and predicting the output

Fig. 5 F1-score and Balanced Accuracy for all Celestrak satellite pairs and for a subset of the Validation synthetic satellite pairs



given the input information. The model performance is analyzed for the two scenarios.

Results show that both scenarios present high and similar performance, with $F1$ and BA of around 88% and 95%, respectively. Results demonstrate that a supervised learning model can be used to predict the encounters in polar, circular, low, and realistic SGP4 orbits only knowing the inclination i , the semi-major axis a , the mean anomaly M , and the right-ascending node Ω . The high performance with SGP4 despite the missing output-dependent information in the input dataset was not expected as the difference between SGP4 and Kepler trajectories within is highly notable. However, based on the results it can be concluded that the satellite-to-satellite distances and the encounter sequences remained nearly invariant over a propagation period of two days.

Nevertheless, a statistical test has revealed that the absence of certain input information in the dataset significantly reduces the performance of the SGP4 model compared to the Kepler model.

The key strengths of our contributions include: (1) a decentralized solution based on synthetic data with strong generalization on realistic data from an available database,

Table 3 Mean model performance obtained for both validation and test sets and for both Kepler and SGP4 scenarios

	Kepler		SGP4	
	Validation	Test	Validation	Test
BA [%]	94.5	96.6	93.4	95.6
$F1$ [%]	88.7	93.1	86.9	91.3
P [%]	90.9	95.1	90.2	94.3
R [%]	89.3	93.5	87.1	91.5

(2) the assumption of a realistic orbital model with the most important perturbations in the LEO space, (3) the easy adaptation to different satellite antenna characteristics, and (4) the application the applicability in heterogeneous environments with varying altitudes and inclinations.

Future work must encompass more challenging frameworks, moving from low, circular, and polar orbits to low orbits with different eccentricities and a broader range of inclinations. We expect that larger training sets and more complex Machine Learning models will be required. In this sense, the computational cost of the model inference should

be analyzed and optimized in the near future to be used in a real application.

Acknowledgements This work was supported and funded by the Spanish Ministry of Economic Affairs and Digital Transformation and the European Union–NextGeneration EU, in the framework of the Recovery Plan, Transformation and Resilience (PRTR) (Call UNICO I+D 5G 2021, ref. number TSI-063000-2021-5-6GSatNet-SS), and by the Government of Catalonia in the scope of the NewSpace Strategy for Catalonia.

Author Contributions EF designed and performed the experiment. EF wrote the manuscript with support from JAR-D-A, FB, and JE. JAR-D-A, FB, and JE supervised the project. All authors discussed the results.

Availability of data and material The data that support the findings of this study are available from the corresponding author, Estel Ferrer, upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no competing interests.

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